

The cost of US foods as related to their nutritive value



Background: Comparisons of the cost of different foods relative to their energy and nutritive value were conducted in the 1800s by the US Department of Agriculture (USDA). Objective: The objective was to reestablish the relations between food cost, energy, and nutrients by using contemporary nutrient composition and food prices data from the USDA. Design: The USDA Food and Nutrient Database for Dietary Studies 1.0 (FNDDS 1.0) and the Center for Nutrition Policy and Promotion food prices database were used for analysis. For 1387 foods, key variables were as follows: energy density (kcal/g), serving size (g), unit price (\$/100 g), serving price (\$/serving), and energy cost (\$/kcal). A regression model tested associations between nutrients and unit price (\$/100 g). Comparisons between food groups were tested by using one-factor analyses of variance. Relations between energy density and price within food groups were tested by using Spearman's correlations. Results: Grains and fats food groups supplied the lowest-cost dietary energy. The energy cost for vegetables was higher than that for any other food group except for fruit. Serving sizes increased with water content and varied inversely with energy density of foods. The highest prices per serving were for meats, poultry, and fish, and the lowest prices per serving were for the fats category. Although carbohydrates, sugar, and fat were associated with lower price per 100 g, protein, fiber, vitamins, and minerals were associated with higher price per 100 g, after adjustment for energy. Conclusions: Grains and sugars food groups were cheaper than vegetables and fruit per calorie and were cheaper than fruit per serving. These price differentials may help to explain why low-cost, energy-dense foods that are nutrient poor are associated with lower education and incomes.